

3.3: Power Functions and Polynomial Functions

Definition. (Power Function)

A **power function** is a function that can be represented in the form

$$f(x) = kx^p$$

where k and p are real numbers, and k is known as the **coefficient**.

Example. Identify which of the following functions are power functions:

$f(x) = 1$

Constant

$f(x) = x$

Identity

$f(x) = x^2$

Quadratic

$f(x) = x^3$

Cubic

$f(x) = \frac{1}{x}$

Reciprocal

$f(x) = \frac{1}{x^2}$

Reciprocal squared

$f(x) = \sqrt{x}$

Square root

$f(x) = \sqrt[3]{x}$

Cube root

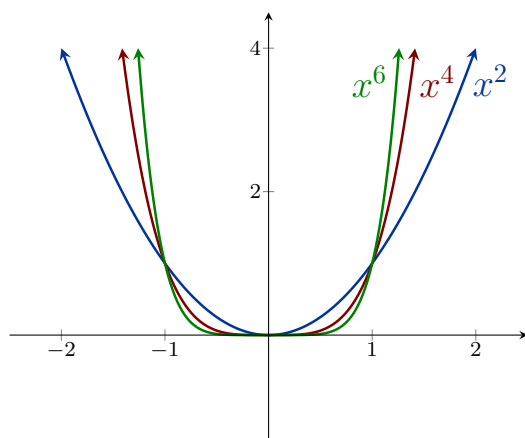
$f(x) = 2^x$

Exponential

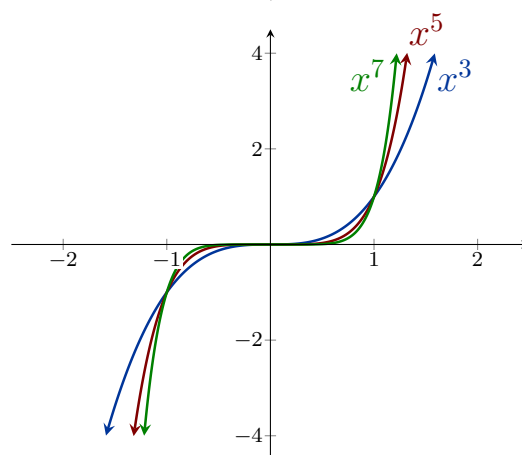
Identifying End Behavior of Power Functions

Below are example graphs of power functions $f(x) = kx^n$:

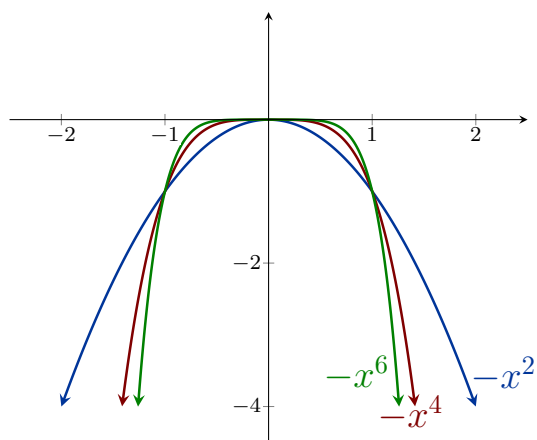
n Even, $k > 0$



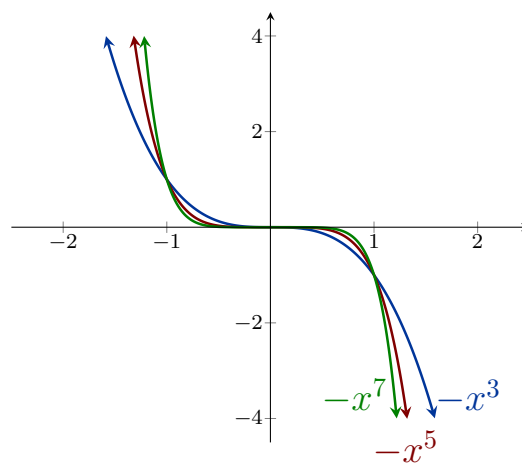
n Odd, $k > 0$



n Even, $k < 0$



n Odd, $k < 0$



Example. Describe the end behavior of the graphs of the following power functions:

$$f(x) = x^8$$

$$g(x) = -x^9$$

$$x \rightarrow -\infty \quad f(x) \rightarrow \infty$$

$$x \rightarrow -\infty \quad g(x) \rightarrow \infty$$

$$x \rightarrow \infty \quad f(x) \rightarrow \infty$$

$$x \rightarrow \infty \quad g(x) \rightarrow -\infty$$

Definition. (Polynomial Functions)

Let n be a non-negative integer. A **polynomial function** is a function that can be written in the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

Each a_i is a coefficient and can be any real number, and $a_n \neq 0$.

The **degree** of the polynomial is the highest power that the variable is raised to. The **leading coefficient** is the coefficient associated with the term containing the highest power of the variable.

Example. Describe the end behavior of the graphs of the following polynomials: [Graph](#)

$$f(x) = 5x^4 + 2x^3 - x - 4$$

Degree 5

$$x \rightarrow -\infty \quad f(x) \rightarrow \infty$$

$$x \rightarrow \infty \quad f(x) \rightarrow \infty$$

$$h(x) = 3x^5 - 4x^4 + 2x^2 + 1$$

Degree 5

$$x \rightarrow -\infty \quad h(x) \rightarrow -\infty$$

$$x \rightarrow \infty \quad h(x) \rightarrow \infty$$

$$g(x) = -2x^6 - x^5 + 3x^4 + x^3$$

Degree 6

$$x \rightarrow -\infty \quad g(x) \rightarrow \infty$$

$$x \rightarrow \infty \quad g(x) \rightarrow \infty$$

$$j(x) = -6x^3 + 7x^2 + 3x + 1$$

Degree 3

$$x \rightarrow -\infty \quad j(x) \rightarrow \infty$$

$$x \rightarrow \infty \quad j(x) \rightarrow -\infty$$

Definition. (Intercepts and Turning Points of Polynomial Functions)

A **turning point** is a point at which the graph changes direction from increasing to decreasing, or decreasing to increasing.

The y -intercept is the point at which the function has an input value of zero.

The x -intercepts are the points at which the output value is zero.

A polynomial of degree n will have, at most, n x -intercepts, and $n - 1$ turning points.

Complex Conjugate Root Theorem

If a polynomial $f(x)$ has real coefficients, and $a + bi$ is a root of $f(x)$, then $a - bi$ is also a root of $f(x)$.

Example. Find the intercepts of

$$f(x) = x^4 - 4x^2 - 45$$

$$\text{Let } y = x^2$$

$$\rightarrow f(x) = y^2 - 4y - 45$$

$$= y^2 - 9y + 5y - 45$$

$$= y(y-9) + 5(y-9)$$

$$= (y+5)(y-9)$$

$$= (x^2+5)(x^2-9)$$

$$\text{solve } f(x) = 0$$

$$(x^2+5)(x^2-9) = 0$$

$$x^2+5=0$$

$$x^2 = -5$$

$$x = \pm \sqrt{5} i$$

$$x^2-9=0$$

$$x^2 = 9$$

$$x = \pm 3$$

$$(-3, 0)$$

$$(3, 0)$$